# Quasi-Newton Methods

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**Notations.** We write $\langle \cdot, \cdot \rangle$ for the Euclidean inner product and $\| \cdot \|$ for its induced norm. Let $\mathbb{S}^n$, $\mathbb{S}^n_+$, and $\mathbb{S}^n_{++}$ denote the real symmetric, positive semidefinite, and positive definite $n \times n$ matrices, respectively. For $A, B \in \mathbb{S}^n$, we write $A \succeq B$ when $A - B \in \mathbb{S}^n_+$.

# 1 Preliminaries

## 1.1 Useful inequalities

**Lemma 1.1** (Arithmetic–geometric mean inequality)**.** *For any integer $K \geq 1$ and nonnegative scalars $r_1, \ldots, r_K$,*

$$\left(\prod_{k=1}^K r_k\right)^{1/K} \leq \frac{1}{K} \sum_{k=1}^K r_k.$$

**Lemma 1.2.** *If $A \in \mathbb{S}^n_{++}$ and $\mu I \preceq A \preceq LI$ for $0 \leq \mu \leq L$, then, for every $x \in \mathbb{R}^n$,*

$$\frac{\mu}{2}\|x\|^2 \leq \frac{1}{2}\langle x, Ax \rangle \leq \frac{L}{2}\|x\|^2.$$

## 1.2 Convexity and smoothness

**Convexity.** A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if

$$f\big(\theta x + (1-\theta)y\big) \leq \theta f(x) + (1-\theta) f(y)$$

for all $x, y \in \mathbb{R}^n$ and $\theta \in [0, 1]$. A useful consequence for differentiable convex functions is the first-order inequality.

**Proposition 1.1.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is differentiable and convex, then*

$$f(x) \geq f(y) + \langle \nabla f(y), x - y \rangle \qquad \textit{for all } x, y \in \mathbb{R}^n.$$

**Smoothness.** A continuously differentiable function $f : \mathbb{R}^n \to \mathbb{R}$ is $L$-smooth if

$$\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\| \qquad \text{for all } x, y \in \mathbb{R}^n.$$

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**Lemma 1.3** (Descent lemma)**.** *If $f : \mathbb{R}^n \to \mathbb{R}$ is $L$-smooth, then*

$$f(x) \le f(y) + \langle \nabla f(y), x - y \rangle + \tfrac{L}{2}\|x - y\|^2 \qquad \textit{for all } x, y \in \mathbb{R}^n.$$

**Descent direction I.** For a differentiable function $f$, a *descent direction* at $x$ is a vector $d$ satisfying

$$\langle d, \nabla f(x) \rangle < 0.$$

For fixed $d$, first-order expansion gives, as $\varepsilon \downarrow 0$,

$$f(x + \varepsilon d) = f(x) + \varepsilon \langle d, \nabla f(x) \rangle + o(\varepsilon).$$

Hence $f(x + \varepsilon d) < f(x)$ for all sufficiently small $\varepsilon > 0$. If $f$ is $L$-smooth with $L > 0$, the descent lemma gives the explicit bound

$$f(x + \varepsilon d) \le f(x) + \varepsilon \langle d, \nabla f(x) \rangle + \tfrac{L\varepsilon^2}{2}\|d\|^2.$$

Taking $\varepsilon = -\langle d, \nabla f(x) \rangle / (L\|d\|^2) > 0$ yields

$$f(x + \varepsilon d) \le f(x) - \tfrac{\langle d, \nabla f(x) \rangle^2}{2L\|d\|^2}.$$

In particular, if $\nabla f(x) \neq 0$ and $d = -\nabla f(x)$, then $\varepsilon = 1/L$ and $f\big(x - \frac{1}{L}\nabla f(x)\big) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$.

**Descent direction II.** Let $f$ be differentiable and convex, and let $x^\star$ be a minimizer. Consider the squared distance to this minimizer,

$$\delta(x) := \tfrac{1}{2}\|x - x^\star\|^2.$$

The function $\delta$ is convex, 1-smooth, and 1-strongly convex, but its gradient $\nabla\delta(x) = x - x^\star$ depends on the unknown minimizer. Nevertheless, convexity of $f$ gives

$$\langle -\nabla f(x), \nabla\delta(x) \rangle \le f(x^\star) - f(x) < 0 \qquad \text{whenever } f(x) > f(x^\star).$$

Thus $-\nabla f(x)$ is also a descent direction for $\delta$ at every nonoptimal $x$. Indeed, for $g = \nabla f(x)$,

$$\delta(x - \eta g) = \delta(x) - \eta \langle g, x - x^\star \rangle + \tfrac{\eta^2}{2}\|g\|^2,$$

so a sufficiently small $\eta > 0$ reduces the distance to $x^\star$.

## 1.3 Proximal interpretation of gradient descent

For a differentiable $f$ and $\eta > 0$, define its first-order model at $x$ by $\ell_x(z) = f(x) + \langle \nabla f(x), z - x \rangle$. The gradient step minimizes this model plus a quadratic penalty:

$$x^+_{\mathrm{GD}} = x - \eta \nabla f(x) = \arg\min_z \big\{ \ell_x(z) + \tfrac{1}{2\eta}\|z - x\|^2 \big\}.$$

For a proper, closed, convex function $f$, the proximal point step instead minimizes the original function plus the same penalty:

$$x^+_{\mathrm{Prox}} = \arg\min_z \big\{ f(z) + \tfrac{1}{2\eta}\|z - x\|^2 \big\}.$$